

CYBERSECURITY RECONNAISSANCE & NETWORK SCANNING ASSESSMENT REPORT

Targets: networkwalks.com (External Footprinting) | Local LAN Subnet (Internal Network Scanning)

Assessment Type: Passive/Active Reconnaissance & Host Discovery

Tools Used: whois, whatweb, nslookup, curl, wafw00f, dnsrecon, Zenmap (Nmap GUI)

Environment: Kali Linux (Phase 1) | Windows PC (Phase 2)

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Classification: For Educational and Authorized Testing Purposes Only

1. Executive Summary

This report presents the combined findings of a two-phase reconnaissance assessment conducted for training purposes. Phase 1 focused on external footprinting of the domain

networkwalks.com, using six built-in Kali Linux tools to gather publicly available information regarding domain ownership, hosting infrastructure, web technologies, DNS configuration, and firewall presence. Phase 2 focused on internal network scanning of a local LAN subnet using Zenmap, the graphical front-end for Nmap, to identify live hosts and visually map the network topology.

Together, these two phases replicate the natural progression of the early stages of a penetration test: first profiling an organization's external, internet-facing presence, then shifting to internal network discovery to identify live systems that could be targeted in later scanning and exploitation phases.

2. Scope of Engagement

Item	Detail
External Target	networkwalks.com
Internal Target	Local LAN subnet
Testing Type	Passive/Active Footprinting (Phase 1); Active Host Discovery (Phase 2)
Authorization	Educational Lab Environment
Tools Permitted	whois, whatweb, nslookup, curl, wafw00f, dnsrecon, Zenmap/Nmap
Exclusions	No exploitation, no intrusive scanning beyond host discovery, no data alteration

3. Objectives

The purpose of this combined assessment was to:

- Identify domain registration and ownership details for the external target.
- Determine the technologies powering the target website.
- Resolve the domain to its corresponding IP address.
- Inspect HTTP response headers for server and infrastructure clues.
- Determine whether a Web Application Firewall is protecting the external target.
- Enumerate all available DNS records associated with the domain.
- Identify all live hosts present on the local LAN subnet.
- Visually document the discovered network topology and preserve it as evidence.

4. Methodology

Phase 1 followed a standard passive information-gathering approach, using tools native to Kali Linux that rely solely on publicly accessible data sources (WHOIS databases, DNS records, and HTTP responses). No unauthorized access, exploitation, or intrusive testing was performed against the external target.

Phase 2 followed an active internal discovery approach using Zenmap's Ping Scan profile (nmap -sn), which sends ICMP/ARP probes to every address in the specified subnet to determine which hosts are currently online, without performing any port scanning or service enumeration.

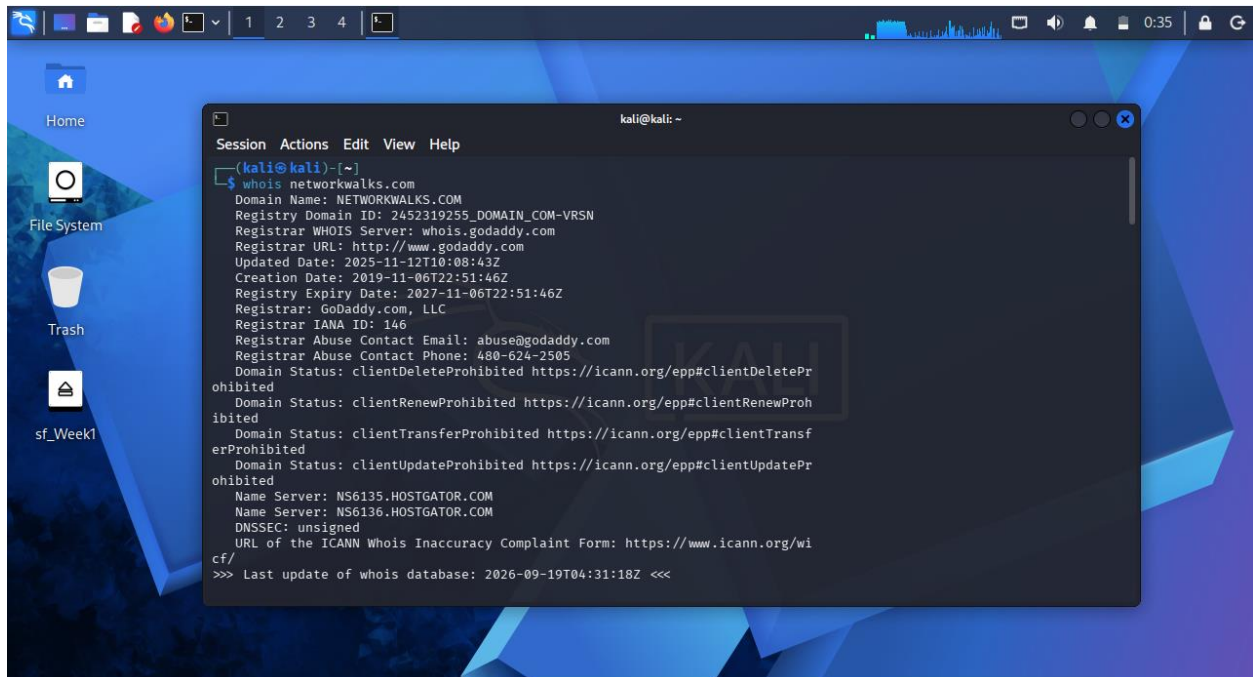
PHASE 1: EXTERNAL FOOTPRINTING — networkwalks.com

5.1 Task 1 — WHOIS Domain Lookup

Objective: Identify domain registration details, ownership, and name servers.

Command Executed:

```
whois networkwalks.com
```

A screenshot of a Kali Linux desktop environment. The desktop background is a blue and purple geometric pattern. On the left side, there are icons for 'Home', 'File System', 'Trash', and 'sf_Week1'. In the center, a terminal window is open, displaying the output of the 'whois networkwalks.com' command. The terminal window has a title bar that says 'kali@kali: ~'. The output shows domain registration details for NETWORKWALKS.COM, including the registrar (GoDaddy.com, LLC), creation date (2019-11-06), expiry date (2027-11-06), and name servers (NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM). The terminal also shows the last update of the whois database as 2026-09-19T04:31:18Z. The desktop taskbar at the bottom shows various application icons and the system clock indicating 0:35.

Observations:

- Registrar: GoDaddy.com, LLC
- Creation Date: 2019-11-06
- Registry Expiry Date: 2027-11-06
- Name Servers: NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM
- Abuse Contact Email: abuse@godaddy.com

Security Implication: The WHOIS record reveals the domain registrar, registration timeline, and hosting provider (HostGator, inferred from the name servers). This information can be leveraged by a malicious actor to identify the hosting environment and to craft social engineering attempts using registrar or abuse contact details.

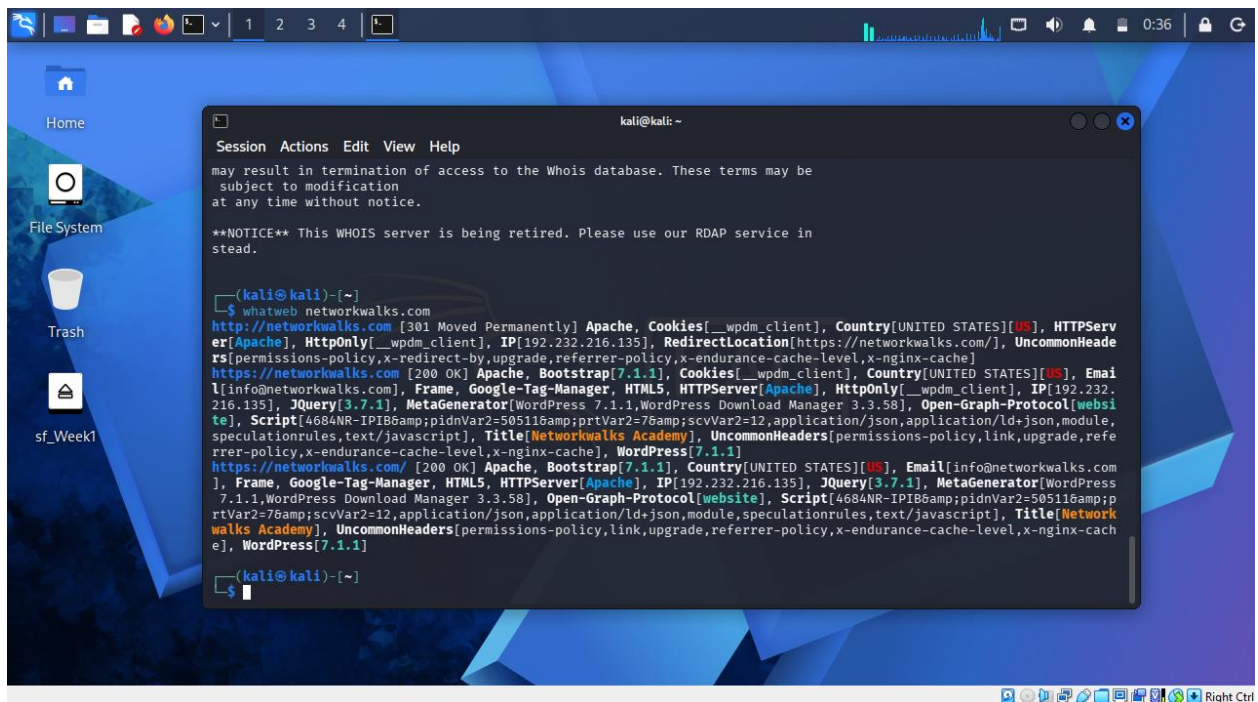
Recommendation: Enable WHOIS privacy/proxy protection through the domain registrar to mask organizational and contact details from public lookup. If privacy protection cannot be enabled due to registrar policy, ensure the abuse contact is a monitored generic mailbox rather than a personal or department-specific address.

5.2 Task 2 — Web Technology Fingerprinting (WhatWeb)

Objective: Identify the web server, CMS, plugins, and frameworks in use.

Command Executed:

```
whatweb networkwalks.com
```



Observations:

- Web Server: Apache
- CMS: WordPress 7.0.4
- Plugin Identified: WP Download Manager 3.3.58
- IP Address: 192.232.216.135
- Additional Metadata: JQuery 3.7.1, Bootstrap 7.0.4, Email address disclosure (info@networkwalks.com)

Security Implication: Disclosure of specific software versions allows an attacker to cross-reference known CVEs to find viable exploitation paths. The exposed email address also increases the risk of targeted phishing.

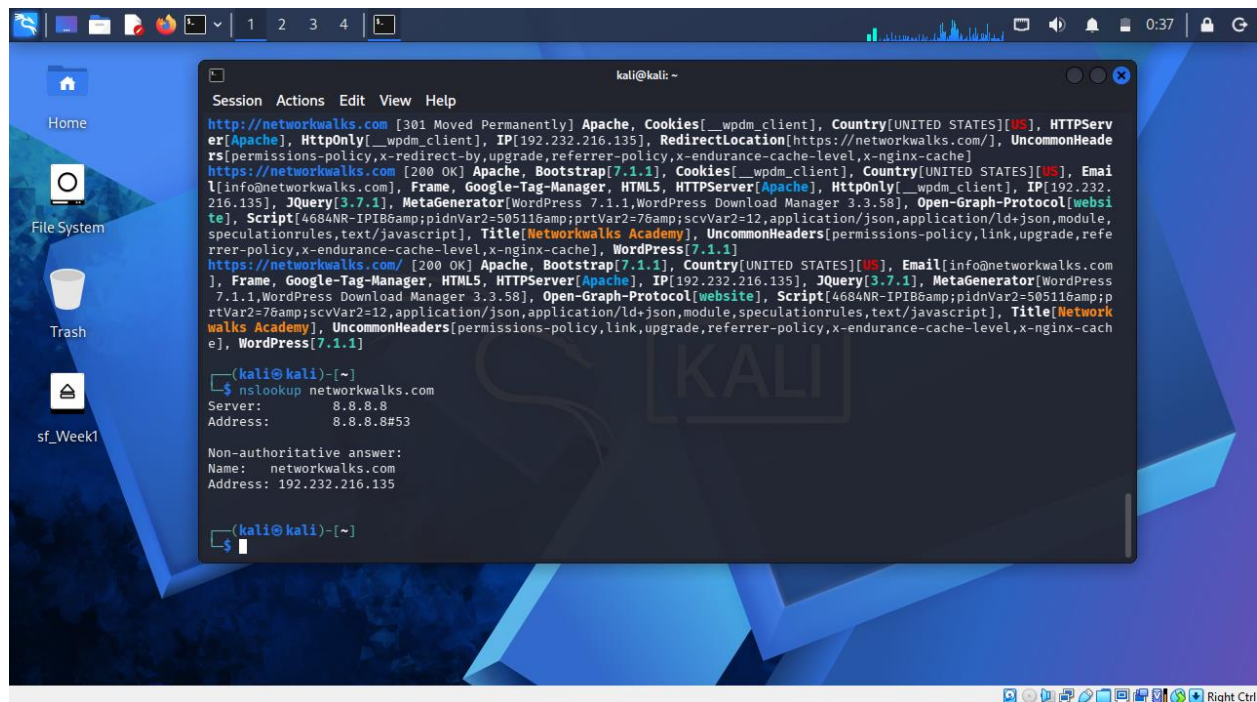
Recommendation: Suppress or randomize version banners for WordPress core and installed plugins. Ensure WordPress core, themes, and the WP Download Manager plugin are kept fully updated. Avoid displaying direct contact emails in plaintext on public-facing pages.

5.3 Task 3 — DNS Resolution (nslookup)

Objective: Resolve the domain name to its corresponding IP address.

Command Executed:

```
nslookup networkwalks.com
```



Observations:

- Resolved IP Address: 192.232.216.135
- DNS Server Queried: 8.8.8.8

Security Implication: Knowledge of the real IP address enables an attacker to conduct direct port scanning and infrastructure mapping, potentially bypassing protections that rely solely on domain-based filtering.

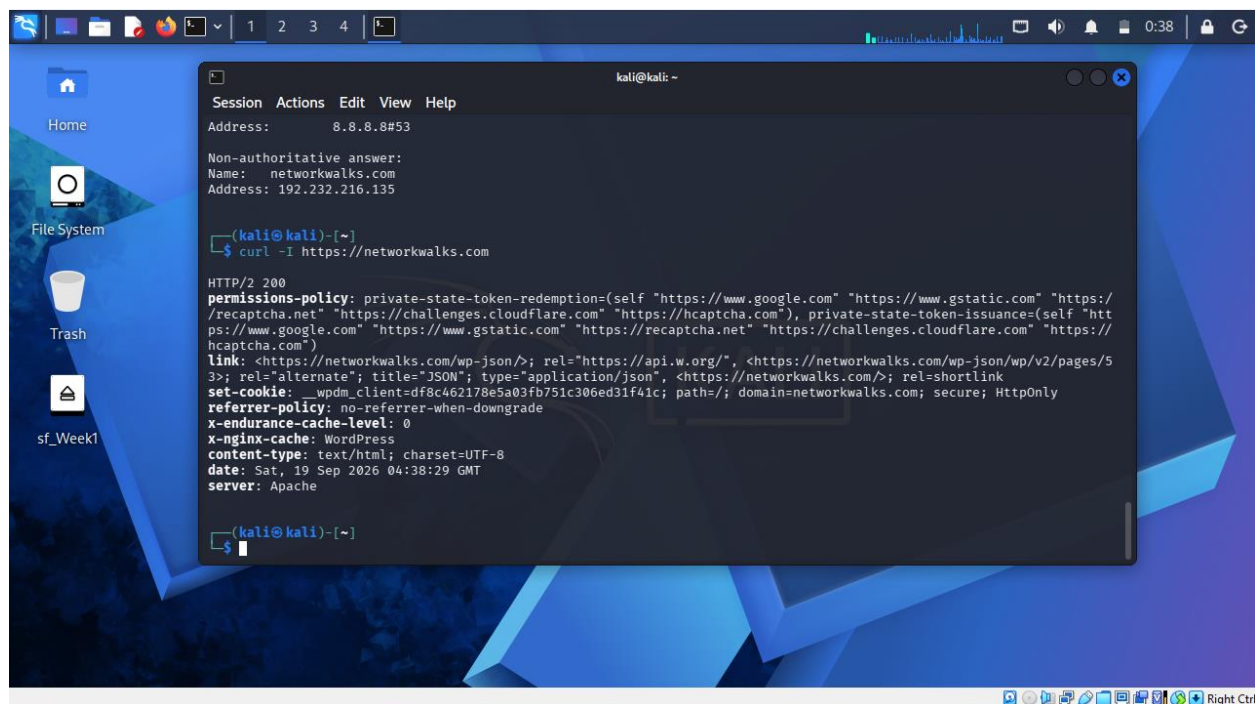
Recommendation: Route all public traffic through a CDN/reverse proxy service (e.g., Cloudflare) that masks the origin server IP. Configure the origin server's firewall to only accept connections from the CDN's known IP ranges.

5.4 Task 4 — HTTP Header Inspection (curl)

Objective: Examine HTTP response headers for server banners, cookies, and redirect behavior.

Command Executed:

```
curl -I https://networkwalks.com
```



Observations:

- HTTP Status: 200 OK (HTTP/2)
- Server: Apache
- Cache Layer: x-nginx-cache: WordPress
- Cookie Disclosed: __wpdm_client (path=/, domain=networkwalks.com; secure; HttpOnly)
- REST API Endpoint Exposed: /wp-json/

Security Implication: The response headers confirm the underlying server software and expose the WordPress REST API endpoint, which can be queried further to enumerate site metadata.

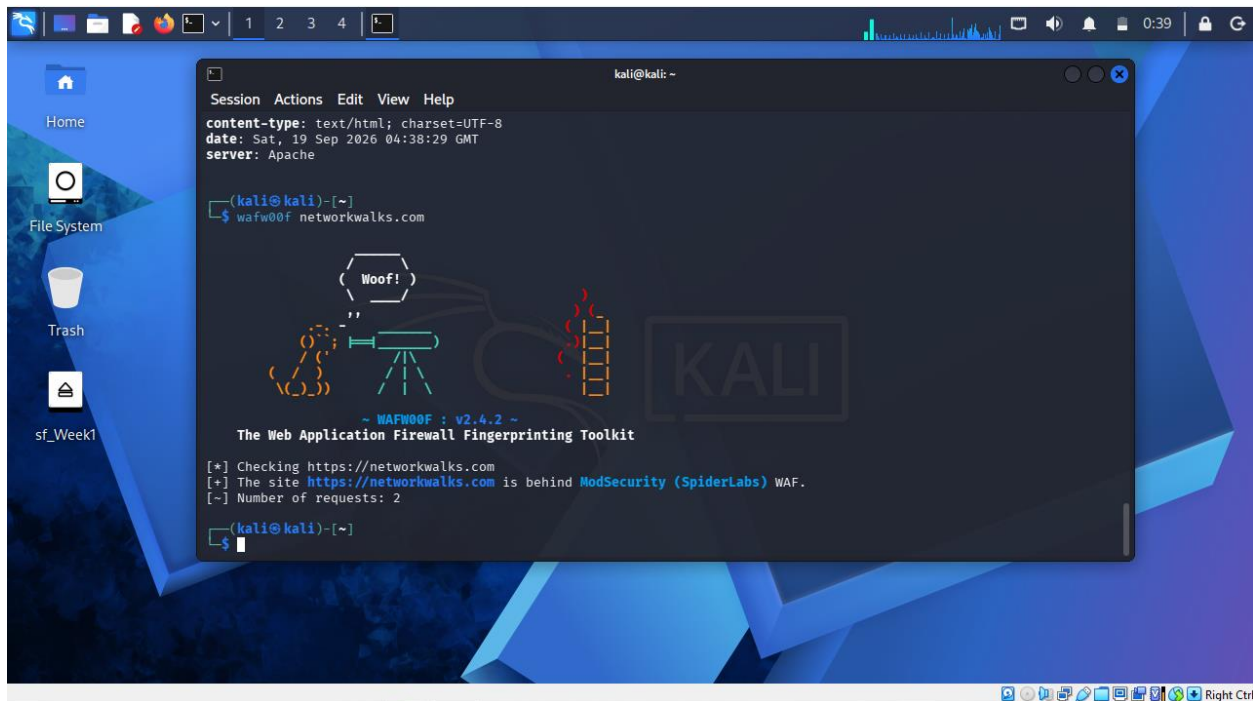
Recommendation: Disable or restrict access to the /wp-json/ REST API endpoint for unauthenticated users. Remove or obscure the Server header value using web server configuration directives (e.g., ServerTokens Prod in Apache).

5.5 Task 5 — Web Application Firewall Detection (wafw00f)

Objective: Determine whether a WAF is actively protecting the target.

Command Executed:

```
wafw00f networkwalks.com
```



Observations:

- WAF Detected: ModSecurity (SpiderLabs)
- Number of Requests Sent: 2

Security Implication: Confirmation of an active WAF informs the attacker that basic attack payloads are likely to be blocked, forcing a more sophisticated, evasive approach in later attack phases.

Recommendation: Continue maintaining and regularly updating the ModSecurity rule set (OWASP Core Rule Set recommended). Periodically test the WAF configuration using authorized penetration testing.

5.6 Task 6 — DNS Enumeration (dnsrecon)

Objective: Enumerate all associated DNS records, including mail servers, SPF, TXT, and SRV records.

Command Executed:

```
dnsrecon -d networkwalks.com
```

A screenshot of a Kali Linux desktop environment. The desktop background is a blue and black abstract design. On the left side, there is a sidebar with icons for Home, File System, Trash, and sf_Week1. The top of the desktop shows a taskbar with various application icons and a system tray on the right with a clock showing 0:41. A terminal window is open in the center, displaying the output of the 'dnsrecon -d networkwalks.com' command. The terminal output shows various DNS records including SOA, NS, MX, A, and SRV records, along with SPF and TXT records. The terminal window has a title bar that says 'kali@kali: ~' and a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'.

Observations:

- SOA Record: ns6135.hostgator.com
- DNS Software Version: BIND 9.16.23-RH
- MX Record: mail.networkwalks.com → 192.232.216.135
- SPF/TXT Record: v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
- SRV Records: 8 cPanel-related autodiscover records identified

Security Implication: The DNS enumeration reveals the mail server configuration, SPF policy, and the presence of a cPanel-managed hosting environment, useful for spoofing/phishing assessment.

Recommendation: Strengthen the SPF record by changing the qualifier from soft-fail (~all) to hard-fail (-all). Implement DKIM and DMARC alongside SPF. Restrict cPanel access to trusted IP ranges and enforce MFA.

PHASE 2: INTERNAL NETWORK SCANNING — Zenmap

5.7 Task 3 — Discovering Live Hosts on the Local Subnet

Objective: Identify the list of live hosts/PCs present within the local LAN subnet using a Ping Scan in Zenmap.

Command Executed:

```
nmap -sn 192.168.100.2/24
```

Procedure: Zenmap was opened, the local subnet (192.168.100.2/24) was entered as the target, and the "Ping Scan" profile was selected before initiating the scan.

Observations:

- **Total Addresses Scanned:** 256 (entire /24 subnet)
- **Live Hosts Identified:** 8
- **Live Host IP Addresses:** 192.168.100.1 (dev.opt), 192.168.100.2, 192.168.100.3, 192.168.100.4, 192.168.100.5, 192.168.100.7, 192.168.100.8, 192.168.100.9, 192.168.100.10
- **Scan Duration:** 7.14 seconds

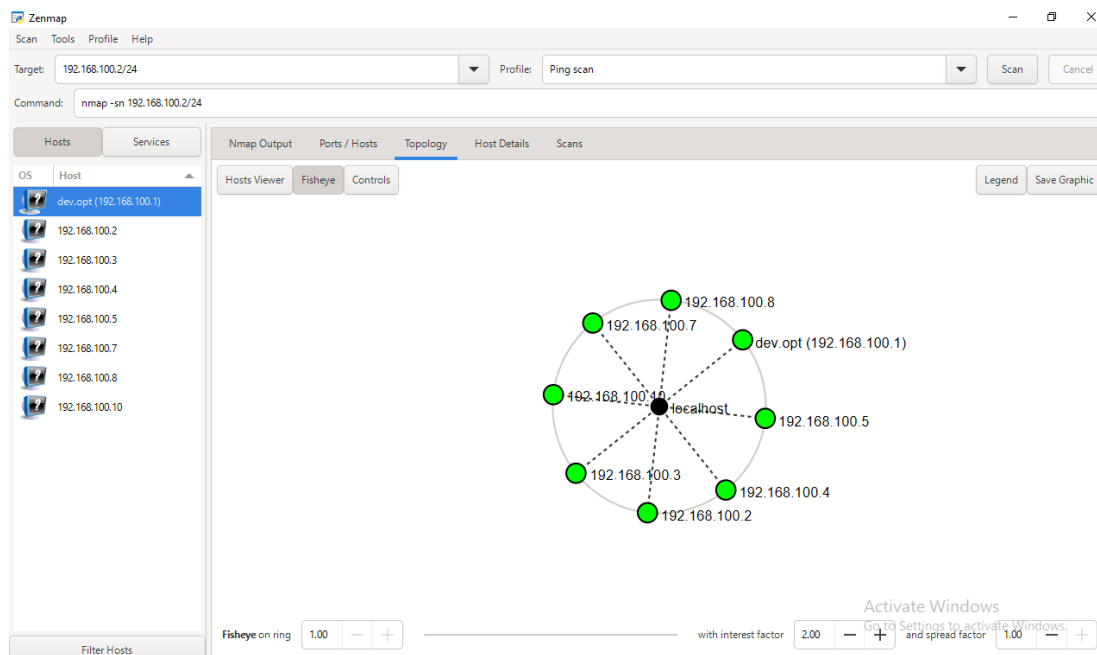
Security Implication: A ping sweep is typically the very first active step an attacker takes after gaining access to a network segment. Identifying live hosts allows an attacker to narrow down further port scanning and enumeration efforts to only active systems, saving time and reducing the number of detectable probes sent across the network. The VMware-prefixed MAC addresses also reveal that the discovered hosts are virtual machines, which may indicate a lab, test, or virtualized production environment.

Recommendation: Configure host-based and network-based firewalls to drop or rate-limit unsolicited ICMP Echo Request (ping) traffic where operationally feasible, making host discovery more difficult for an unauthorized user on the network. Deploy network segmentation (VLANs) to limit the number of hosts visible within a single broadcast domain, reducing the impact of a successful ping sweep. Monitor for unusual volumes of ICMP or ARP traffic using an IDS/IPS, as this is a common indicator of active network reconnaissance.

5.8 Task 7 — Documenting Network Topology (PDF Export)

Objective: Display and save the visual network topology generated by Zenmap in PDF format for reporting and evidence purposes.

Procedure: Within Zenmap, the "Topology" tab was selected following the completed ping scan. The topology legend was enabled to interpret host icons and connection types, and the topology graphic was exported using the "Save Graphic" function, selecting PDF as the output file format.



Observations:

- **Topology View:** Displayed a central "**localhost**" node connected directly to **8 discovered hosts** in a radial/fisheye layout
- **Discovered Hosts Shown:** 192.168.100.1 (dev.opt), 192.168.100.2, 192.168.100.3, 192.168.100.4, 192.168.100.5, 192.168.100.7, 192.168.100.8, 192.168.100.10
- **Host Status Indicators:** All discovered hosts were represented as **green nodes**, indicating live hosts with fewer than three open ports at this stage (ping scan only, no port scanning performed yet)
- **View Mode Active:** Fisheye view was selected, with **Fisheye on ring: 1.00** and **interest factor: 2.00, spread factor: 1.00**
- **Output Option Available:** A "**Save Graphic**" button was visible in the top-right corner of the topology panel for exporting the topology view.

Security Implication: Visual topology mapping allows both attackers and defenders to quickly understand the structure and scale of a network segment. For an attacker, this provides an at-a-glance view of potential pivot points and target density. For a defender or tester, this visual documentation serves as critical evidence for reporting and as a baseline for detecting unauthorized devices introduced into the network at a later date.

Recommendation: Maintain an up-to-date, authorized network topology diagram and regularly compare it against live scan results to detect rogue or unauthorized devices (a practice known as network baselining). Restrict the ability to run network discovery tools (such as Nmap/Zenmap)

to authorized administrative personnel only, using endpoint controls where appropriate. Retain topology exports as part of standard change-management and audit documentation.

Attached Evidence: [Insert exported topology PDF file here, or reference as Appendix A]

6. Consolidated Summary of Findings

Phase	Task	Tool	Key Information Discovered	Risk Level
1	Task 1	whois	Registrar, hosting provider, registration dates	Low–Medium
1	Task 2	whatweb	CMS (WordPress 7.0.4), plugin version, IP, email	Medium–High
1	Task 3	nslookup	Public IP address (192.232.216.135)	Medium
1	Task 4	curl -I	Server banner, cache system, exposed API endpoint	Medium
1	Task 5	wafw00f	WAF identified (ModSecurity – SpiderLabs)	Informational
1	Task 6	dnsrecon	Mail servers, SPF policy, cPanel SRV records	Medium
2	Task 3	Zenmap (Ping Scan)	8 live hosts identified with IP/MAC addresses	Medium
2	Task 7	Zenmap (Topology Export)	Visual network map documented and exported as PDF	Informational

7. Risk Analysis

The combined findings from both phases illustrate how an attacker (or an authorized tester) can progressively build a complete picture of an organization's environment — starting from purely public, internet-facing data (Phase 1) and moving into internal, network-level discovery once access to the local segment is available (Phase 2).

The external footprinting phase revealed a moderate level of information disclosure typical of small-to-medium business websites, including CMS version details and DNS/mail configuration. The internal scanning phase confirmed that basic host discovery via ICMP is currently unrestricted within the tested subnet, allowing rapid enumeration of all active devices. While no critical vulnerabilities were directly identified, the aggregation of this information significantly reduces the effort required for subsequent scanning, enumeration, and potential exploitation phases.

8. Consolidated Recommendations

Priority	Recommendation	Related Task
High	Update WordPress core and WP Download Manager plugin to the latest version	Phase 1 – Task 2
High	Restrict/disable unauthenticated access to /wp-json/ REST API	Phase 1 – Task 4
High	Mask origin server IP using a CDN/reverse proxy (e.g., Cloudflare)	Phase 1 – Task 3
Medium	Enable WHOIS privacy protection	Phase 1 – Task 1
Medium	Implement DKIM and DMARC alongside existing SPF record	Phase 1 – Task 6
Medium	Restrict or rate-limit ICMP ping responses across the internal network	Phase 2 – Task 3
Medium	Implement network segmentation (VLANs) to limit host visibility	Phase 2 – Task 3
Medium	Suppress server and software version banners (Apache, WordPress)	Phase 1 – Task 2, Task 4
Low	Maintain and regularly update an authorized network topology baseline	Phase 2 – Task 7
Low	Regularly update and test ModSecurity WAF rule sets	Phase 1 – Task 5
Low	Restrict use of network scanning tools to authorized personnel only	Phase 2 – Task 3, Task 7

9. Conclusion

This combined assessment successfully demonstrated both the external footprint and internal network exposure of the assessed environment. Phase 1 confirmed that publicly available information — domain registration data, DNS records, HTTP headers, and web technology fingerprints — can be aggregated to build a detailed external profile of the target. Phase 2 confirmed that, absent restrictive controls, live hosts within a local subnet can be rapidly and comprehensively enumerated using basic ping sweep techniques, with the resulting topology easily documented and exported for further analysis.

Implementing the recommendations outlined in Section 8 would meaningfully reduce the exposed attack surface at both the external and internal levels. These findings collectively serve as the foundation for the next phase of this assessment: detailed port and service scanning of the identified live hosts.

10. Legal & Ethical Disclaimer

This assessment was conducted strictly within an authorized educational lab environment for training purposes. All external reconnaissance activities were limited to passive, publicly available information gathering. All internal network scanning activities were limited to non-

intrusive host discovery (ping scan) performed on a private lab subnet owned and controlled by the assessor. No unauthorized access, exploitation, or disruption of services was attempted at any stage of this engagement. This report and its contents must not be used to conduct unauthorized testing against any system or network without explicit written permission from the system/network owner.

End of Report